Year 12 Further Maths - Poisson distributions

Similar questions - Rates, sums and modes

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In the manufacture of fibre optic cable (FOC), flaws occur randomly. Whether any point on a cable is flawed is independent of whether any other point is flawed. The number of flaws in 100 m of FOC of standard diameter is denoted by X, and X has the distribution $\text{Po}(1.8)$.

(a) State a further assumption needed for X to be well modelled by a Poisson distribution. **[1]**

(b) Find the probability that in 300 m of FOC of standard diameter there are exactly 3 flaws. **[2]**

The number of flaws in 100 m of FOC of a larger diameter has the distribution $\text{Po}(1.1)$.

(c) Find the probability that in 100 m of FOC of standard diameter and 100 m of FOC of the larger diameter the total number of flaws is at least 4. **[3]**

2 On any day, the number of orders received in one randomly chosen hour by an online supplier can be modelled by the distribution $\text{Po}(90)$.

(a) Find the probability that at least 20 orders are received in a randomly chosen 10-minute period. **[2]**

(b) Find the probability that in a randomly chosen 10-minute period on one day and a randomly chosen 10-minute period on the next day a total of at least 40 orders are received. **[3]**

(c) State a necessary assumption for the validity of your calculation in part (b). **[1]**

3

(a) A substance emits particles randomly at a constant average rate of 2.4 per minute. A second substance emits particles randomly, and independently of the first source, at a constant average rate of 3.9 per minute. Find the probability that the total number of particles emitted by the two sources in a five-minute period is less than 30. **[3]**

(b) The random variable X represents the number of particles emitted by a substance in a fixed time interval of t minutes. In general the rate at which particles are emitted is proportional to the mass of the substance, but each emission reduces the mass. Explain why a Poisson distribution may not be a valid model for X if t is very large. **[1]**

(c) The random variable Y has the distribution $\text{Po}(\lambda)$. It is given that $\text{P}(Y = r) = \text{P}(Y = r+1)$ and $\text{P}(Y = r) = 1.25\,\text{P}(Y = r-1)$. Determine the value of r and the value of λ. **[6]**

4 The manager of an emergency hotline believes that calls are made to the hotline independently and at a constant average rate throughout the day. From a small random sample of the population, the manager finds that the mean number of calls made in a 1-hour period is 9.6. Let R denote the number of calls made in a randomly chosen 1-hour period.

(a) Using evidence from the small sample, state a suitable distribution with which to model R. **[1]**

(b) In this part use the distribution from part (a).

(i) Find $\text{P}(R > 12)$. **[2]**

(ii) Given that $\text{P}(R = r) > \text{P}(R = r+1)$, show algebraically that $r > 8.6$. **[2]**

(iii) Hence write down the mode of the distribution. **[1]**

(c) The manager also finds, from records over many years, that the modal value of R is 7. Use this result to comment on the validity of the distribution used in part (a). **[1]**

(d) Assume that the distribution used in part (a) is valid, with unknown mean. Find the range of values of the parameter that would correspond to the modal value of R being 7. **[2]**

5 A radar device detects flaws in a motorway before they become dangerous. The number of flaws in a 1 km stretch of motorway is denoted by X. It may be assumed that flaws occur randomly.

(a) State two further assumptions that are necessary for X to be well modelled by a Poisson distribution. **[2]**

Assume that X can be modelled by the distribution $\mathrm{Po}(4.2)$.

(b) Determine the probability that in a randomly chosen stretch of motorway of length 1 km there are between 5 and 8 flaws inclusive. **[2]**

(c) Determine the probability that in two randomly chosen, non-overlapping stretches of motorway, each of length 4 km, there are at least 20 flaws in one stretch and fewer than 15 flaws in the other stretch. **[3]**

6 In this question you must show detailed reasoning. The random variables X and Y denote the number of telephone calls and the number of e-mails, respectively, received by a company in a randomly chosen one-minute period in a working day. For any one-minute period, the following assumptions can be made.

- Calls are received randomly and independently of one another, at a constant average rate.
- E-mails are received randomly and independently of one another, at a constant average rate.
- Calls and e-mails are received independently of one another.

It is known that $\mathrm{E}(X) = 3$.

(a) Find the probability that 4 calls are received in a randomly chosen one-minute period. **[1]**

(b) A sample of 10 independent observations of X is obtained. Find the expected number of these observations in the interval $2 < X < 8$. **[3]**

It is also known that $\mathrm{P}(X+Y=4) = \frac{128}{27}\mathrm{P}(X=2)\mathrm{P}(Y=2)$.

(c) Determine the possible values of $\mathrm{E}(Y)$. **[7]**

(d) Explain where in your solution to part (c) you have used the assumption that calls and e-mails are received independently. **[1]**

Question 1

(a) Flaws occur at a constant average rate (uniformly) along the cable

(b) 300 m: Po(5.4), $\mathrm{P}(=3) = \mathrm{e}^{-5.4}\dfrac{5.4^3}{3!} = 0.1185$

(c) Total $\sim$ Po(2.9). $\mathrm{P}(\geqslant 4) = 1-(0.0550+0.1596+0.2314+0.2237) = 0.330$

Question 2

(a) 10 minutes: Po(15). $\mathrm{P}(\geqslant 20) = 1-\mathrm{P}(\leqslant 19) = 0.125$

(b) Total $\sim$ Po(30). $\mathrm{P}(\geqslant 40) = 1-\mathrm{P}(\leqslant 39) = 0.0463$

(c) The numbers of orders in the two periods are independent (or the rate is the same on both days)

Question 3

(a) Total in 5 minutes $\sim$ Po$(5(2.4+3.9)) =$ Po(31.5). $\mathrm{P}(\leqslant 29) = 0.371$

(b) For large t the mass (so the rate of emission) decreases significantly, so the rate is no longer constant

(c) $\dfrac{\mathrm{e}^{-\lambda}\lambda^r}{r!} = \dfrac{\mathrm{e}^{-\lambda}\lambda^{r+1}}{(r+1)!}$ gives $\lambda = r+1$. $\dfrac{\mathrm{P}(Y=r)}{\mathrm{P}(Y=r-1)} = \dfrac{\lambda}{r} = 1.25$, so $r+1 = 1.25r$ and $r=4$, $\lambda = 5$ (check: $\mathrm{P}(4) = \mathrm{P}(5) = 0.1755$)

Question 4

(a) $R \sim$ Po(9.6)

(b)(i) $\mathrm{P}(R>12) = 1-\mathrm{P}(R\leqslant 12) = 0.172$

(ii) $\dfrac{\mathrm{e}^{-9.6}9.6^r}{r!} > \dfrac{\mathrm{e}^{-9.6}9.6^{r+1}}{(r+1)!}$ gives $1 > \dfrac{9.6}{r+1}$, so $r > 8.6$

(iii) The probabilities decrease from $r=9$ onwards (as $r \geqslant 9 > 8.6$) and are increasing before, so the mode is 9

(c) A modal value of 7 differs from the predicted mode 9, suggesting that the Poisson model with mean 9.6 may not be valid (or the rate is not constant through the day)

(d) Mode 7 requires $\mathrm{P}(7) > \mathrm{P}(8)$, i.e. $\lambda < 8$, and $\mathrm{P}(7) > \mathrm{P}(6)$, i.e. $\lambda > 7$. So $7 < \lambda < 8$

Question 5

(a) Flaws occur independently of one another; flaws occur at a constant average rate (singly)

(b) $\mathrm{P}(5\leqslant X\leqslant 8) = \mathrm{P}(X\leqslant 8)-\mathrm{P}(X\leqslant 4) = 0.382$

(c) 4 km: Po(16.8). $\mathrm{P}(\geqslant 20) = 0.2476$, $\mathrm{P}(\leqslant 14) = 0.2971$. Two possible stretches: $2(0.2476)(0.2971) = 0.147$

Question 6

(a) $X \sim$ Po(3), $\mathrm{P}(X=4) = \mathrm{e}^{-3}\dfrac{3^4}{4!} = 0.168$

(b) $\mathrm{P}(2<X<8) = \mathrm{P}(3\leqslant X\leqslant 7) = 0.5649$. Expected number $= 10(0.5649) = 5.65$

(c) Let $\mathrm{E}(Y) = \mu$. $X+Y \sim$ Po$(3+\mu)$. $\dfrac{\mathrm{e}^{-(3+\mu)}(3+\mu)^4/24}{\mathrm{e}^{-3}\frac{9}{2}\cdot\mathrm{e}^{-\mu}\frac{\mu^2}{2}} = \dfrac{128}{27}$ gives $\dfrac{(3+\mu)^4}{54\mu^2} = \dfrac{128}{27}$, so $(3+\mu)^4 = 256\mu^2$ and $(3+\mu)^2 = 16\mu$ (taking positive roots). So $\mu^2-10\mu+9=0$, $(\mu-1)(\mu-9)=0$. $\mathrm{E}(Y) = 1$ or 9

(d) In writing $\mathrm{P}(X+Y=4)$ as the Poisson probability with mean $3+\mu$ and in using $\mathrm{P}(X=2,Y=2) = \mathrm{P}(X=2)\mathrm{P}(Y=2)$